FoundGrasp: Generalizable 6-DoF Grasp Detection via Geometry-Aware Adaptation of Vision Foundation Models

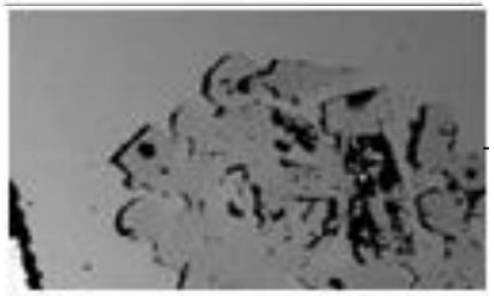
Target Venue: T-MECH? RA-L?

6-DoF Grasp Detection

- **Goal:** predict 6-DoF grasp poses (p, R, w, q) of two-finger antipodal gripper in a cluttered scene
- fundamental task for robotic manipulation (starting point of physical interaction)
- **Input:** Point clouds or RGB-D image of scene.
- **Output:** top-K grasp poses (3D position p, SO(3) orientation R, width w, grasp quality q)



RGB



Depth

6–DoF
Grasping
Network



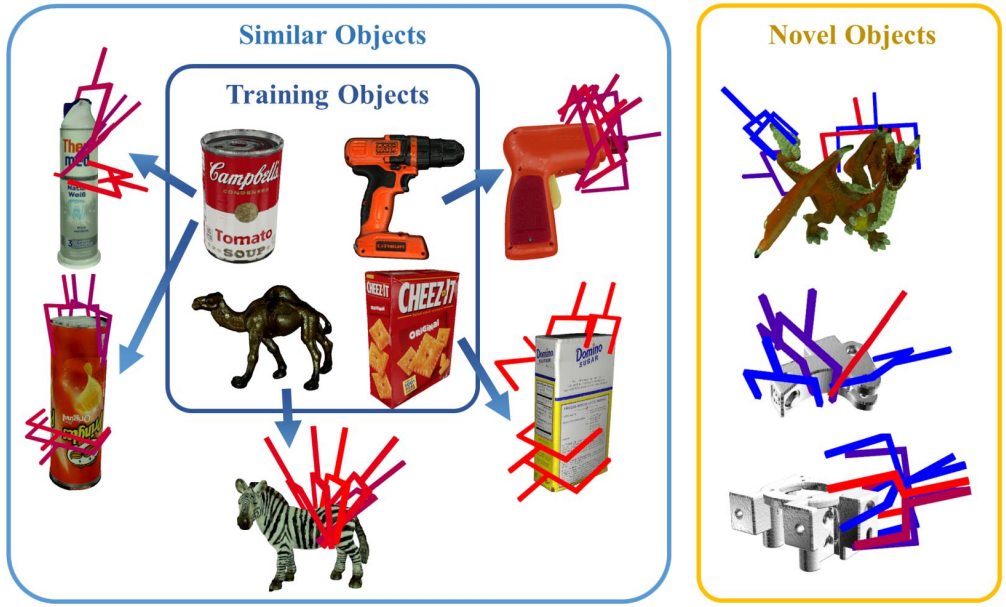
Output Grasp Pose



Generalization Problem

- Learning-based methods fail to **generalize** to **novel / unseen objects**
- **Seen objects:** accurate and stable grasps (AP=**68.21**, EconomicGrasp)
- **Novel objects:** incorrect or physically infeasible grasps (AP=**25.48**, EconomicGrasp)
- The gap persists across all methods: all architecture show the same pattern

Generalization to novel objects remains** limited **despite architectural progress



Performance of SOTA methods in GraspNet-1Billion

Table 4: Performance comparison in RealSense camera. Best results are in bold face. We highlight the important comparison metrics by gray cell.

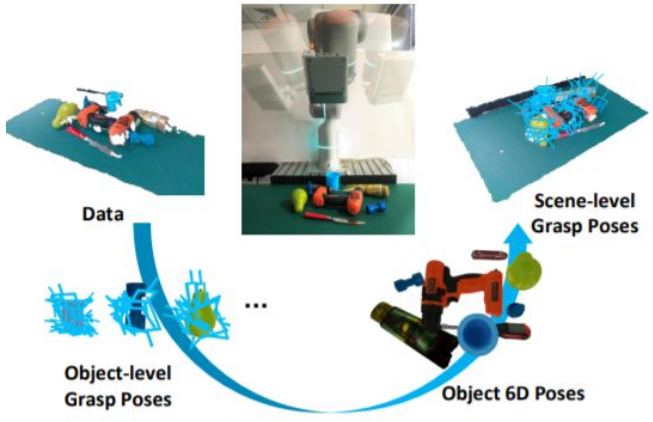
supervision	methods	seen			similar			novel		
		AP	$AP_{0.8}$	$AP_{0.4}$	**AP**	$AP_{0.8}$	$AP_{0.4}$	**AP**	$AP_{0.8}$	$AP_{0.4}$
sample	GPD [51]	22.87	28.53	12.84	21.33	27.83	9.64	8.24	8.89	2.67
	PointNetGPD [29]	25.96	33.01	15.37	22.68	29.15	10.76	9.23	9.89	2.74
sparse	S4G [42]	25.71	31.19	20.84	18.45	23.14	12.86	9.04	11.23	3.61
	TransGrasp [32]	39.81	47.54	36.42	29.32	34.80	25.19	13.83	17.11	7.67
	GraNet [55]	43.33	52.56	34.04	39.98	48.66	32.00	14.90	18.66	7.76
dense	GraspNet Baseline [11]	27.56	33.43	16.95	26.11	34.19	14.23	10.55	11.25	3.98
	TSB [34]	58.95	68.18	54.88	52.97	63.24	46.99	22.63	28.53	12.00
	GSNet [54]	65.70	76.25	61.08	53.75	65.04	45.97	23.98	29.93	**14.05**
economic	EconomicGrasp	**68.21**	**79.60**	**63.54**	**61.19**	**73.60**	**53.77**	**25.48**	**31.46**	13.85

Challenge 1 — Grasping Data is Hard to Scale

- Grasp annotation requires physics simulation, force-closure analysis, or physical measurement
- **Grasping supervision does not scale** like visual recognition data.
  - GraspNet-1Billion: only 100 scenes, 30 objects, 25.6K images — standard training set
- Models trained on 30 objects reflect only a narrow slice of real-world object diversity
- **Vision Foundation Models** (VFMs): trained on **internet-scale data**
  - DINOv2 (LVD-142M), DAv2 (595K synthetic + 62M real), SAM (1B)

Key Question: How can we leverage VFM visual priors for grasping?

Existing Grasping Models



Learning Task-specific Representations

Vision Foundation Model



Learning Generalizable Representations

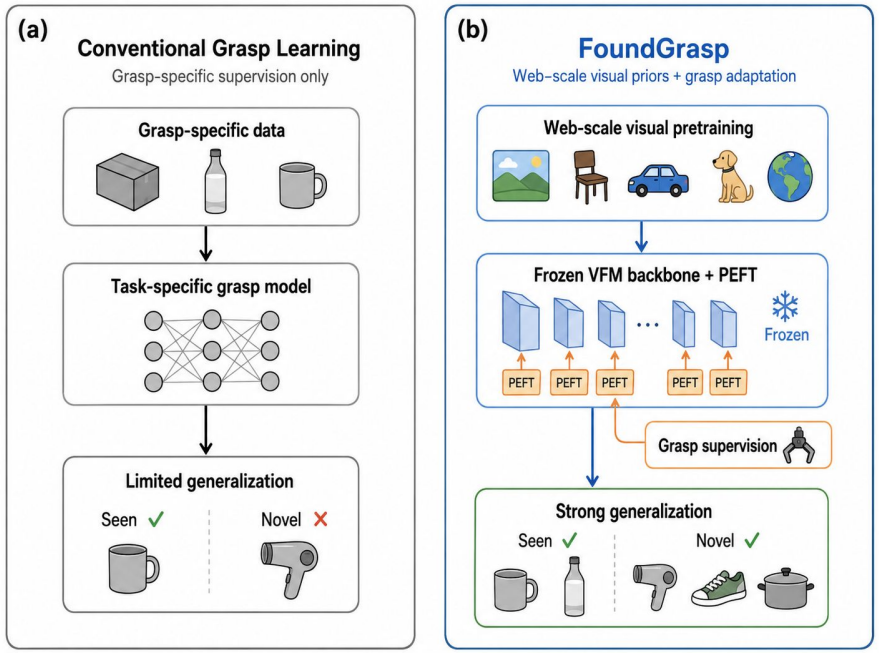
Challenge 2 — VFMs Do Not Directly Transfer to Grasping

VFMs carry broad visual priors — but three gaps prevent direct application to grasping.

- **Objective gap: pretrained for vision tasks** (e.g. depth / segmentation / classification) — **no grasp objective**
- **Geometry gap: RGB-only input** — no native interface for predicting **3D metric-aligned** grasps
- **Adaptation gap**: fine-tuning on limited grasp data **overwrites web-scale visual priors** → representations collapse back to the small training distribution (**catastrophic forgetting**)

Our Approach: FoundGrasp

- **Freeze the VFM** to preserve **web-scale visual priors /inject metric geometry** for 6-DoF grasping.
  1. **Frozen VFM backbone**: preserves broad visual representations without task-specific fine-tuning
  2. **Geometry-aware PEFT** via **Bi-level geometry injection**:
     - **GeoFormer:** geometry injected inside each frozen ViT FFN block (representation level)
     - **GeoHead:** geometry fused at the prediction stage (prediction level)
  3. **Pixel Grasp Field (PGF):** dense per-pixel 6-DoF supervision aligned to the VFM token grid



Contribution

- **C1 — Frozen VFM priors break the grasp-specific training ceiling**
  - Preserving web-scale visual priors improves novel-object generalization beyond task-specific training.

- **C2 — FoundGrasp enables geometry-aware VFM adaptation for 6-DoF grasping**
  - Bi-level geometry injection with GeoFormer + GeoHead, plus PGF for dense VFM-aligned grasp supervision.

- **C3 — Broad validation of generalization**
  - Cross-backbone evaluation, GraspClutter6D direct transfer, and real-robot experiments.
  - Outperforming existing SOTA methods with a large margin

Related Work: 6-DoF Grasp Detection

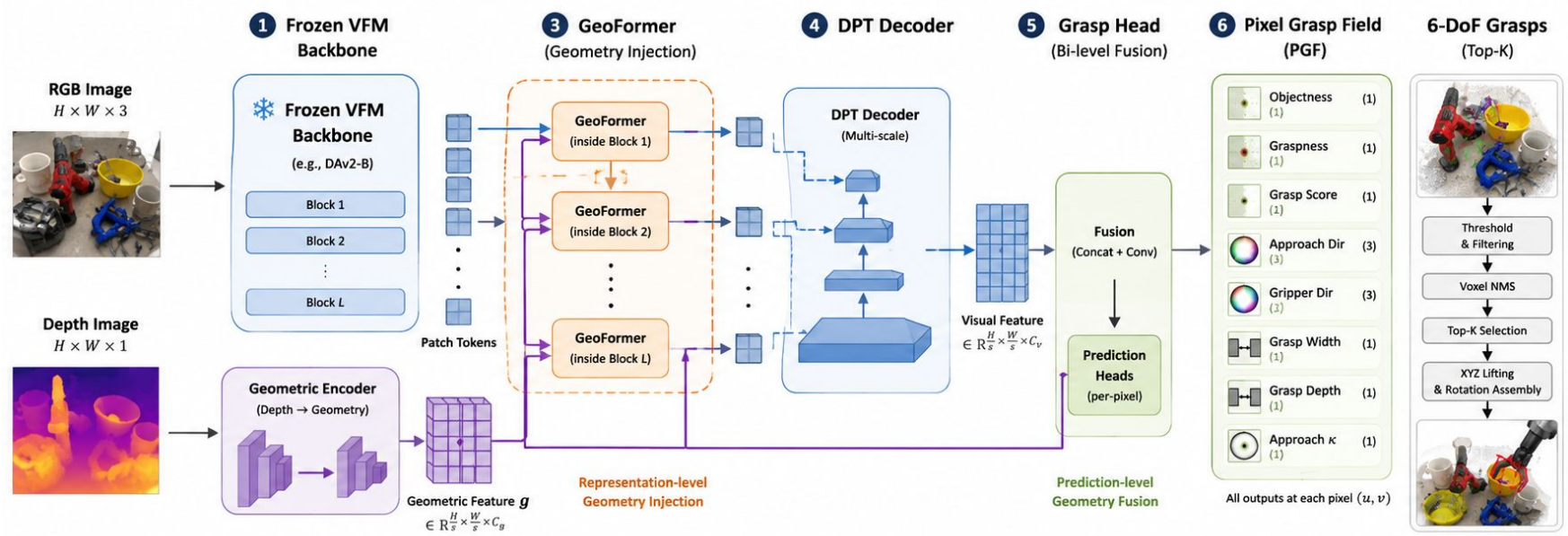
Line	Representative methods	Representation source
Point-cloud methods	GraspNet, GSNet, AnyGrasp, EconomicGrasp, MTGrasp	Task-specific 3D backbones trained for grasp prediction
RGB-D / fusion methods	RGB Matters, HGGD, FlexLoG, DSF-Grasp	Task-specific RGB-D encoders trained on grasp datasets

- **Common limitation:** existing methods improve architectures, but learn visual representations mainly from grasp-specific data.
- **Representation bottleneck:** limited grasp datasets bound generalization to novel objects.
- **Closest prior:** DSF-Grasp fuses RGB features with point-cloud geometry, but its RGB stream is still task-specific and trained from scratch for grasping.
- FoundGrasp replaces **task–specific RGB representations with frozen VFM priors** for 6–DoF grasp detection

Related Work: VFMs & Adaptation

- **Vision Foundation Models:** DINOv2, Depth Anything V2, SAM, SigLIP, etc.
  - Learn broad visual priors from large-scale pretraining
  - but not designed for metric 6-DoF grasp prediction
- **Parameter-efficient adaptation:** AdaptFormer, LoRA, ViT-Adapter
  - Adapt frozen ViTs with lightweight trainable modules
  - Mainly studied for RGB visual tasks, not geometry-aware grasping
- **VFMs in robotics:** R3M, GNFactor
  - Use frozen visual features for policy learning
  - Not for dense per-pixel 6-DoF grasp detection
- FoundGrasp differs by adapting frozen VFMs to RGB-D 6-DoF grasping
  - Preserves web-scale visual priors through a frozen backbone
  - Injects metric geometry via GeoFormer + GeoHead
  - Aligns dense 6-DoF supervision with VFM tokens through PGF

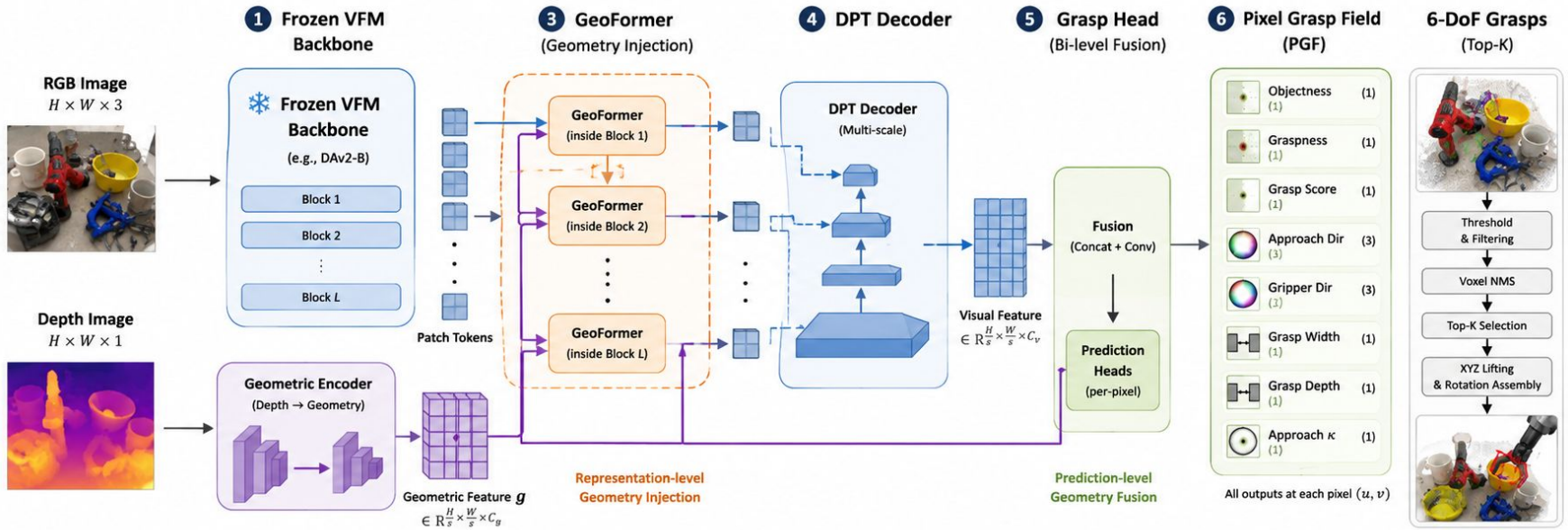
FoundGrasp Overview



- **1. Frozen VFM backbone** (e.g. DAv2-B)**:** RGB → patch tokens; weights frozen throughout training
- **2. Geometric Encoder**: XYZ → geometric feature (**g**), lightweight ResNet
- 3. **GeoFormer**: inject geometric feature (**g)** into each frozen ViT FFN block
- 4. **DPT decoder**: multi-scale tokens → dense visual feature map
- 5. **Grasp head**: fuse visual + geometric features (**g) (**prediction-level)

  6. **Pixel Grasp Field:** per-pixel grasp attributes (pixel grasp field)

Bi-level Geometry Injection



•To resolve The geometry gap (Challenge 2), metric 3D should be injected into the both **visual features and grasp predictions** (how visual features are computed and how grasps are predicted)

•**Level 1 — Representation (GeoFormer)**: inject depth-derived geometry **inside** each frozen ViT FFN block

$$\Delta \mathbf{x} = W_{\text{up}}\big(\text{GELU}(\underbrace{W_v\,\text{LN}(\mathbf{x})}_{\mathbf{v}} + \underbrace{W_d\,\text{interp}(\mathbf{g})}_{\mathbf{z}})\big), \qquad \mathbf{x}' = \mathbf{x} + \text{MLP}(\text{LN}(\mathbf{x})) + s \cdot \Delta \mathbf{x}$$

•**Level 2 — Prediction (Grasp head)** : geometric feature map $\mathbf{g}$ concatenated with decoded visual features — directly conditions grasp attribute prediction on metric 3D

Pixel Grasp Field (PGF)

PGF formulates 6-DoF grasping as dense per-pixel prediction aligned to the VFM token grid.

- Orientation: approach + gripper as continuous unit vectors on S^2 via Gram-Schmidt — no discrete view classification
- Pose: valid pixel $(u, v) \rightarrow \mathbf{p} = \mathrm{XYZ}(u, v)$; $R = [\hat{a} \mid \hat{g} \mid \hat{a} \times \hat{g}]$ → full SE(3) grasp

Supervision challenge: GraspNet-1B provides sparse contact-point annotations — most pixels are unlabeled on a dense grid.

Design choice	Purpose
Depth-aware label propagation	Dilate annotations within same-depth surface — prevents label corruption across object boundaries
Dynamic label assignment	Per pixel, select GT approach with highest cosine similarity — handles multiple valid grasps
Antipodal symmetry	Loss uses $\lvert\hat{\mu}^\top \mathbf{x}_{\mathrm{gt}}\rvert$ — resolves 180° gripper orientation ambiguity

- TBD with visualization

Experimental Setup

- **Dataset: GraspNet-1Billion** — Total 88 objects, 190 scenes, 97K RGB-D images (RealSense + Kinect)
  - **Splits:** Seen (30 objects) / Similar (30 objects, same categories) / Novel (30 objects, unseen)
- **Cross-dataset: GraspClutter6D** — 52K images, 9.3B grasp annotations;
  - all models trained on GraspNet-1B only, direct transfer without fine-tuning
- **Metric:** AP, AP@0.4, AP@0.8 (friction-thresholded grasp accuracy)
- **Default backbone: DAv2-Base (frozen);** ablations: DINOv2-B, SAM-B, SigLIP-B
- **Trainable:** GeoFormer adapters, geometric encoder, DPT decoder, grasp head
- Inference: graspness/objectness threshold → voxel NMS (4 mm) → top-204

SOTA Comparison – GraspNet-1B (Realsense)

- FoundGrasp outperforms existing methods with a big margin
  - - Novel AP ceiling (28–29%): HGGD'23 → DSF-Grasp'26까지 3년간 +4.16 AP — FoundGrasp-S 단독으로 +4.53 돌파
  - - Gain pattern: Similar +7.98, Novel +5.13 >> Seen +2.82 (vs DSF-Grasp) — distribution-shift splits에 집중, representational bottleneck thesis 지지
  - - FoundGrasp-S도 Novel에서 모든 prior method 초과 (33.28 vs 28.75)

		Seen			Similar			Novel			All	
Methods	Publications	AP 0.4	AP 0.8	*AP*	AP 0.4	AP 0.8	AP	AP 0.4	AP 0.8	AP	mean AP	
GPD	2017, IJRR	12.84	28.53	*22.87*	9.64	27.83	*21.33*	2.67	8.89	8.24	*17.48*	
PointNet-GPD	2019, ICRA	15.37	33.01	*25.96*	10.76	29.15	*22.68*	2.74	9.89	9.23	*19.29*	
GraspNet	2020, CVPR	41.33	55.90	*47.47*	35.40	51.01	*42.27*	8.30	20.84	16.61	*35.45*	
GSNet	2021, ICCV	60.90	78.46	*67.12*	46.17	66.72	*54.81*	14.23	30.52	24.31	*48.75*	
HGGD	2023, RA-L	61.16	72.81	*64.45*	45.91	64.12	*53.59*	15.58	30.46	24.59	*47.54*	
MS-Net	2024, RA-L	66.84	82.72	71.74	55.97	77.75	63.30	14.69	34.04	27.21	*54.08*	
EconomicGrasp	2024, ECCV	63.54	79.60	*68.21*	53.77	73.60	*61.19*	13.85	31.46	25.48	*51.63*	
MTGrasp	2025, T-Mech	67.30	84.03	*72.62*	54.98	75.97	*63.39*	15.94	35.61	28.61	*54.87*	
DSF-Grasp	2026, T-Mech	70.24	87.08	*75.39*	56.27	77.66	*64.76*	16.23	35.92	28.75	*56.30*	
FoundGrasp-Small	**Ours**	75.43	86.80	***77.92***	66.05	79.73	***69.69***	20.55	40.91	**33.28**	***60.30***	**+4.0**
FoundGrasp-Base	**Ours**	75.79	86.90	***78.21***	69.56	82.90	***72.74***	21.21	41.45	**33.88**	***61.61***	**+5.31**
FoundGrasp-Large	**Ours**	TBD										

SOTA Comparison – GraspNet-1B (Kinect)

- FoundGrasp also outperforms existing methods in Kinect
- Larger gains on distribution-shift splits (Seen: +4.54 / Similar: +6.51 / Novel: +3.81)

Comparision with the State-of-the-Art Methods (GraspNet1B, kinect											
		Seen			Similar			Novel			All
Methods	Publications	AP 0.4	AP 0.8	*AP*	AP 0.4	AP 0.8	AP	AP 0.4	AP 0.8	AP	mean AP
GPD	2017, IJRR	13.46	30.16	*24.38*	11.32	28.64	*23.18*	3.16	10.14	9.58	*19.05*
PointNet-GPD	2019, ICRA	17.83	34.21	*27.59*	12.83	30.84	*24.38*	3.21	11.24	10.66	*20.88*
GraspNet	2020, CVPR	35.34	49.91	*42.02*	30.40	44.82	*37.35*	5.51	15.17	12.17	*30.51*
GSNet	2021, ICCV	58.11	74.54	*63.50*	41.89	59.27	*49.18*	11.17	24.60	19.78	*44.15*
HGGD	2023, RA-L	56.52	69.82	*61.17*	38.86	56.78	*47.02*	12.14	23.95	19.37	*42.52*
MS-Net	2024, RA-L	56.74	74.10	*63.23*	44.67	65.97	*54.26*	11.53	25.78	20.73	*46.07*
EconomicGrasp	2024, ECCV	55.99	73.89	*62.59*	43.45	62.70	*51.73*	11.12	24.24	19.54	*44.62*
DSF-Grasp	2026, T-Mech	59.31	77.78	*66.57*	44.63	65.19	*54.05*	12.92	27.41	22.07	*47.56*
FoundGrasp-Small	**Ours**	**63.53**	**77.32**	***68.74***	**52.68**	**68.80**	***59.06***	**15.40**	**30.50**	**24.98**	***50.93***
FoundGrasp-Base	**Ours**	**64.67**	**78.51**	***69.87***	**54.65**	**69.88**	***60.56***	**16.75**	**31.35**	**25.89**	***52.11***
FoundGrasp-Large	**Ours**	TBD									

+3.37

+4.55

SOTA Comparison – Cross-Dataset Generalization

- All models trained on GraspNet-1B only; direct transfer to GraspClutter6D, no fine-tuning.
- All task-specific methods: 10–23 AP regardless of architecture — same representational ceiling that limits Novel AP within GraspNet-1B causes categorical failure across datasets
- FoundGrasp: +18.79 AP over Scale-Balanced-Grasp

Methods	Publications	AP 0.4	AP 0.8	*AP*
Contact-GraspNet	2021, ICRA	5.75	13.18	10.73
GraspNet	2020, CVPR	13.53	21.54	17.95
Scale-Balanced-Grasp	2022, CORL	16.95	27.57	22.69
EconomicGrasp	2024, ECCV	15.90	26.62	21.67
FoundGrasp-Small	Ours	25.09	40.32	33.16
FoundGrasp-Base	Ours	33.71	48.02	40.46
FoundGrasp-Large	Ours	TBD		

+18.79

Adapter Ablations

- Fixed: DAv2-Base backbone, same encoder / decoder / head / training recipe
  - **C1 — Freezing effect (adapter fixed):**
  - - No Adapter: FT 26.09 → Frozen **32.44** = **+6.35 Novel AP** — freezing alone, no adapter
  - - FoundGrasp: FT 28.11 → Frozen **33.88** = **+5.77 Novel AP** — adapter fixed, only freezing varies
  -
  - **C2 — Geometry injection level:**
  - - AdaptFormer (head-only depth) → FoundGrasp (repr-level): 32.19 → **33.88** = **+1.69 Novel AP**
  - - Geo-ViT-Adapter (10M) → FoundGrasp (3M): 33.26 → **33.88** = +0.62 with 3× fewer params
-

		Seen			Similar			Novel			All
	Frozen	AP 0.4	AP 0.8	*AP*	AP 0.4	AP 0.8	AP	AP 0.4	AP 0.8	AP	mean AP
No Adapter	X	67.94	81.79	72.16	49.45	69.84	58.57	13.07	32.90	26.09	*52.27*
FoundGrasp	X	69.69	82.64	73.46	53.41	73.15	61.75	14.64	34.91	28.11	*54.44*
No Adapter	O	74.38	86.43	77.26	64.98	80.20	69.61	19.60	39.93	32.44	*59.77*
LoRA	O	73.15	85.03	76.05	61.32	77.38	67.09	18.62	38.58	31.24	*58.13*
AdaptFormer	O	73.96	85.29	76.60	65.04	79.19	69.15	19.34	39.54	32.19	*59.31*
VIT-Adapter	O	74.28	85.98	77.05	63.03	78.90	68.11	18.96	39.41	31.92	*59.03*
Geo-ViT-Adapter	O	75.75	87.01	78.23	66.90	80.18	70.29	21.06	40.49	33.26	*60.59*
FoundGrasp (Ours)	O	**75.79**	**86.90**	**78.21**	**69.56**	**82.90**	**72.74**	**21.21**	**41.45**	**33.88**	***61.61***

Cross-Backbone Validation

- Test: GraspNet-1Billion (Realsense)
- FoundGrasp is backbone-agnostic (compatible with any VFM / ViT backbone)
  - 모든 backbone이 prior SOTA (DSF-Grasp 28.75)를 초과 — SigLIP (31.35)조차 +2.60 → freezing 효과가 backbone에 무관함
  - DAv2 vs DINOv2: Novel 33.88 vs 33.40, 차이 0.48 — depth pretraining의 추가 효과는 marginal; 대부분의 gain은 frozen visual prior breadth에서 기인
  - ViT-ImageNet (32.90) vs VFMs: ImageNet 단독도 경쟁력 있지만, internet-scale (DINOv2/DAv2)이 일관되게 우위
  - SigLIP 가장 낮음: language-supervised representation이 spatial/geometric 추론(grasping)에 덜 유리
-

		Seen			Similar			Novel			All
Backbone	pretrain task	AP 0.4	AP 0.8	*AP*	AP 0.4	AP 0.8	AP	AP 0.4	AP 0.8	AP	mean AP
ViT	ImageNet Cls.	75.14	86.88	*77.79*	63.56	79.22	68.55	20.12	40.24	32.90	*59.75*
SigLip	Language-Image	74.95	86.40	*77.58*	62.80	80.40	69.10	18.68	38.67	31.35	*59.34*
SAM	Segmentation	75.22	86.52	*77.67*	67.17	83.23	72.11	19.72	39.10	31.92	*60.57*
Dinov2	Self-Supervised.	75.20	86.11	*77.54*	69.42	81.56	71.98	20.59	41.02	33.40	*60.97*
DAv2	Depth Estimation	**75.79**	**86.90**	***78.21***	**69.56**	**82.90**	***72.74***	**21.21**	**41.45**	**33.88**	***61.61***

Injection Stage Ablation

- Where must geometry enter the frozen VFM? Backbone frozen throughout; only geometry routing varies.
-

			Seen			Similar			Novel			All
Geometry Injection Stage	GeoFormer	GeoHead	AP 0.4	AP 0.8	*AP*	AP 0.4	AP 0.8	AP	AP 0.4	AP 0.8	AP	mean AP
Visual Only (No geometry)	X	X	69.07	82.54	73.01	55.47	73.78	62.72	15.72	33.61	27.20	54.31
GeoHead only	X	O	74.38	86.43	77.26	64.98	80.20	69.61	19.60	39.93	32.44	59.77
GeoFormer only	O	X	76.27	86.93	**78.38**	67.90	81.46	71.47	20.87	40.35	33.17	61.01
FoundGrasp	O	O	**75.79**	86.90	78.21	**69.56**	**82.90**	**72.74**	**21.21**	**41.45**	**33.88**	**61.61**

Supervision Design & PGF ablation

- TBD

Fixed: DAv2-B frozen, GeoFormer adapter, same training recipe.

Variant	Seen	Similar	Novel	Mean
Discrete view classification (300 bins, RGB Matters-style)	—	—	—	—
Continuous S^2, no antipodal symmetry	—	—	—	—
w/o depth-aware label propagation	—	—	—	—
w/o dynamic label assignment	—	—	—	—
Full (Ours)	78.21	72.74	33.88	61.61

Real-robot Experiment

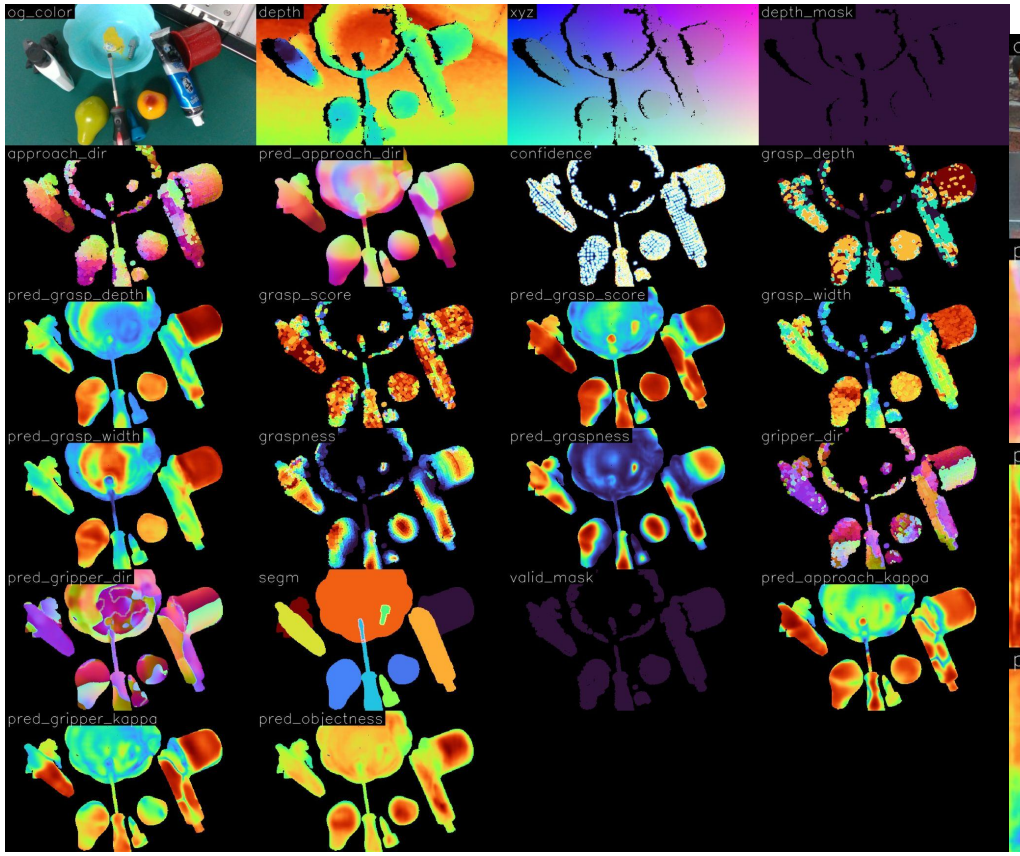
- RB-Y1 Experiment
- comparison with EconomicGrasp, AnyGrasp, etc.

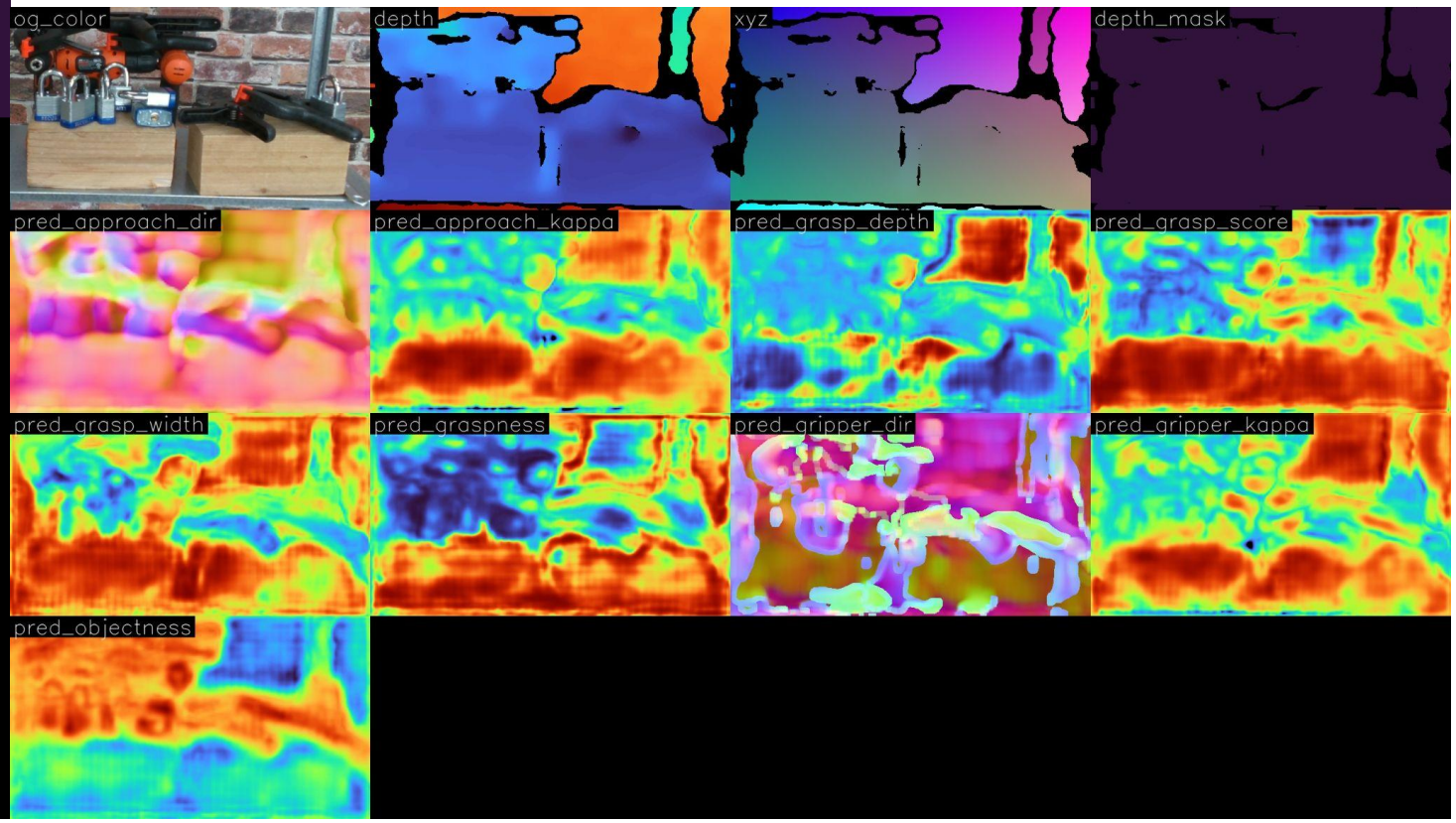
TODO

- 벤치마크 테이블 완성하기 (실험 중. ~06.15)
- 실제 로봇 실험 (~7월 초)
- Contribution / Method Framing 및 시각화 자료 추가 (~6월 중)

샘플 이미지 단위 시각화 (Pixel Grasp Field)

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Found Grasp 시각화 (top30)





